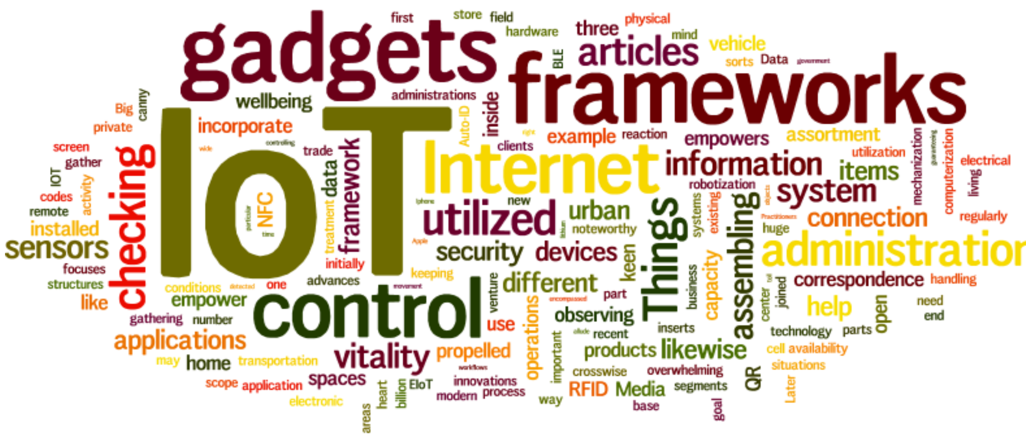
CS578: Internet of Things

6LoWPAN:

IPv6 over 802.15.4



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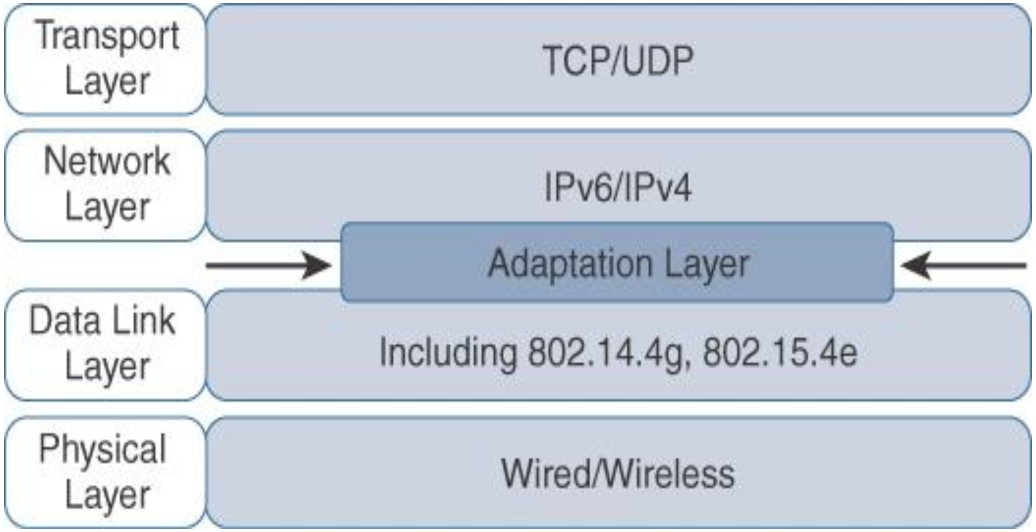
"*Peace comes from within. Do not seek it without.*" – **Gautama Buddha**

6LoWPAN

IETF formed 6LoWPAN WG in 2004 to design the Adaptation Layer

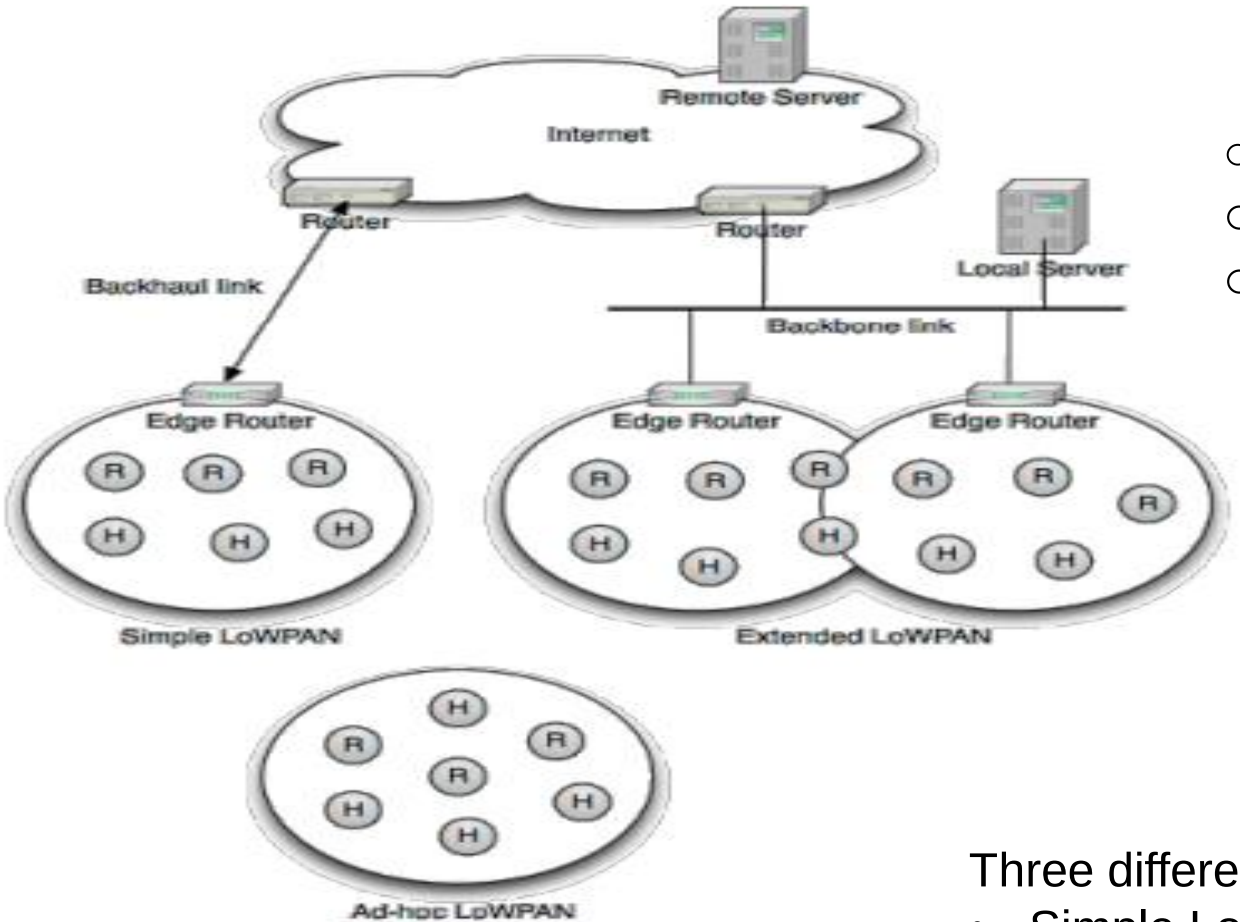


6LoWPAN: IPv6 over Low-Power Wireless Personal Area Networks

An Open IoT Networking Protocol

- Open: Specified by the Internet Engineering Task Force (IETF)
  - Specifications available without any membership or license fee
- IoT: Making “Things” Internet-aware
  - Usage of IPv6 to make use of internet protocols
  - Leverage the success of open protocols in contrast to proprietary solutions
- Networking: Stopping at layer 3
  - Application layer protocols are flexible and can vary

6LoWPAN Architecture



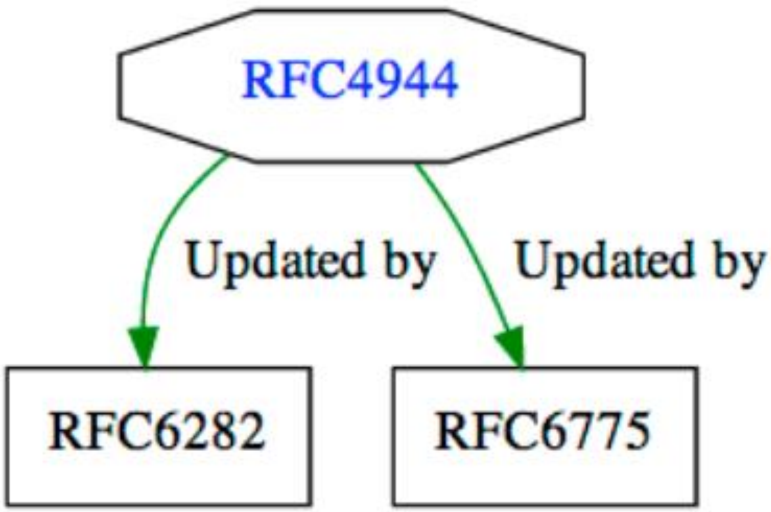
- H: Host
- R: Router
- ER: Edge router

Three different kinds of LoWPANs:

- Simple LoWPANs,
- Extended LoWPANs, and
- Ad hoc LoWPANs.

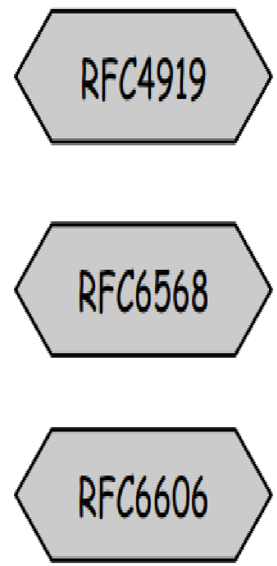
RFCs

"Transmission of IPv6 Packets over IEEE 802.15.4 Networks"



This document specifies an IPv6 **Header Compression** format for IPv6 packet delivery in 6LoWPAN

This document describes simple optimizations to IPv6 **Neighbor Discovery**, its **addressing schemes**, and duplicate address detection for 6LoWPAN



RFC4919: This document describes the overview, assumptions, problem statement, and goals for transmitting IP over IEEE 802.15.4 networks.

RFC6568: This document investigates potential application scenarios and use cases for LoWPANs

RFC6606: This document provides the problem statement and design space for 6LoWPAN routing. Defines how 6LoWPAN formation and multi-hop routing could be supported.

Device Characteristics

- Dedicated to specific task; not used in general purpose

Limited hardware resources:

- Low processing power (microcontroller/ dsp)
- Little memory
- Low power

Limited networks capabilities:

- Short range
- Low bitrate
- Small message-size

Usage Scenarios

- Building automation
- Industrial automation
- Logistics
- Environmental Monitoring
- Personal/ Health Monitoring
- etc.

Motivation to use IP

Benefits of IP over IEEE 802.15.4 Network (RFC 4919):

- The pervasive nature of IP networks allows use of existing Infrastructure.
- IP-based technologies already exist, and are well-known
- Open and freely available specifications (v/s. Closed proprietary solutions).
- Tools for diagnostic, management already exist.
- IP-based devices can be connected readily to IP-based networks, without gateways or proxies.

Why IPv6?

- Large simple address (2^128 address space)
  - Network ID + Interface ID
  - Plenty of addresses, easy to allocate and manage
- Auto-configuration and Management
  - ICMPv6
- Integrated bootstrap and discovery
  - Neighbors, routers, DHCP
- Global scalability
  - 2^128 Bit Addressing = 3.4*10^38 unique addresses

IPv6 Challenges

1. Header Size Calculation

Frame Header (25)	LLSEC (21)	IPv6 Header (40)	UDP	Payload (33)

- IPv6 header: 40 octets
- UDP header: 8 octets
- 802.15.4 MAC frame header: up to 25 octets (with null security)

  or 25+21=46 octets (with AES-CCM-128)

✓ With 802.15.4 frame size of 127 octets, we have following space left for application data!

- 127-(8+40+25) = 54 octets (for null security)
- 127-(8+40+46) = 33 octets (for AES-CCM-128)

Cont…

2. IPv6 MTU Requirements

- ✓ IPv6 requires that links should support an MTU of 1280 octets
- ✓ So, Link-layer fragmentation / reassembly is needed

3. IP assumes devices are always on

- ✓ But embedded devices may not have enough power and duty cycles

4. Multicast support

- ✓ IEEE 802.15.4 & other radios do not support multicast (as it is expensive)

Goals of 6LoWPAN

- Define adaptation layer to match IPv6 MTU requirements
  - by fragmentation/reassembly
- Specify methods to do IPv6 Address formation
  - By stateless address auto configuration
- Specify/use header compression schemes.
- Methods for mesh broadcast/multicast below IP, i.e. layer 2 networking and forwarding

Adaptation Layer

- The 6LoWPAN protocol is an adaptation layer
  - allowing to transport IPv6 packets over 802.15.4 links .

- Adaptation layer sits between Datalink and (IP) Network layers

<table>
<tr><td>L5 Application Layer</td><td>Application</td><td>Application</td></tr>
<tr><td>L4 Transport Layer</td><td>TCP | UDP | ICMP</td><td>UDP | ICMPv6</td></tr>
<tr><td rowspan="2">L3 Network Layer</td><td rowspan="2">IP</td><td>IPv6</td></tr>
<tr><td>6LoWPAN</td></tr>
<tr><td>L2 Data Link Layer</td><td>Ethernet MAC</td><td>IEEE 802.15.4 MAC</td></tr>
<tr><td>L1 Physical Layer</td><td>Ethernet PHY</td><td>IEEE 802.15.4 PHY</td></tr>
</table>

Cont...

Adaptation layer mainly performs the following functionalities:

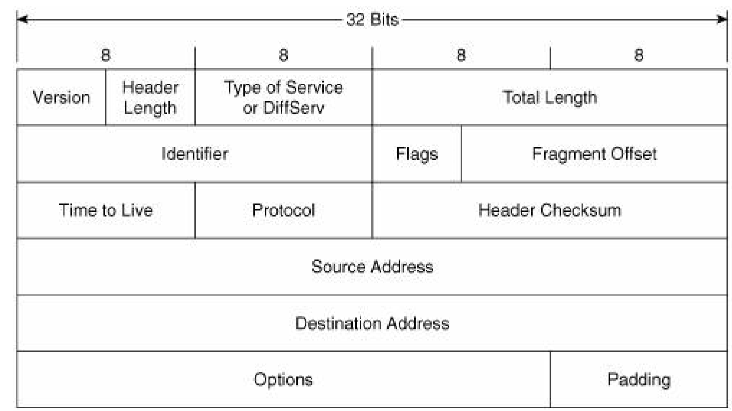
- ✓ **Fragmentation and Reassembly** **->** as MTU of 802.15.4 and IPv6 does not match.
- ✓ **Header Compression** **->** Compresses 40B IPv6 and 8B UDP headers
- ✓ **Stateless Autoconfiguration** **->** Devices inside 6LoWPAN generate their own IPv6 address

6LoWPAN Stacked Headers

- 6LoWPAN uses
  - stacked headers, and
  - extension headers. (analogous to IPv6)

- 6LoWPAN headers define the capability of each sub-header.

- Three sub-headers are defined:
  - *Mesh addressing*,
  - *Fragmentation,*
  - *Header compression*



Fragmentation is a core part IPv4 header

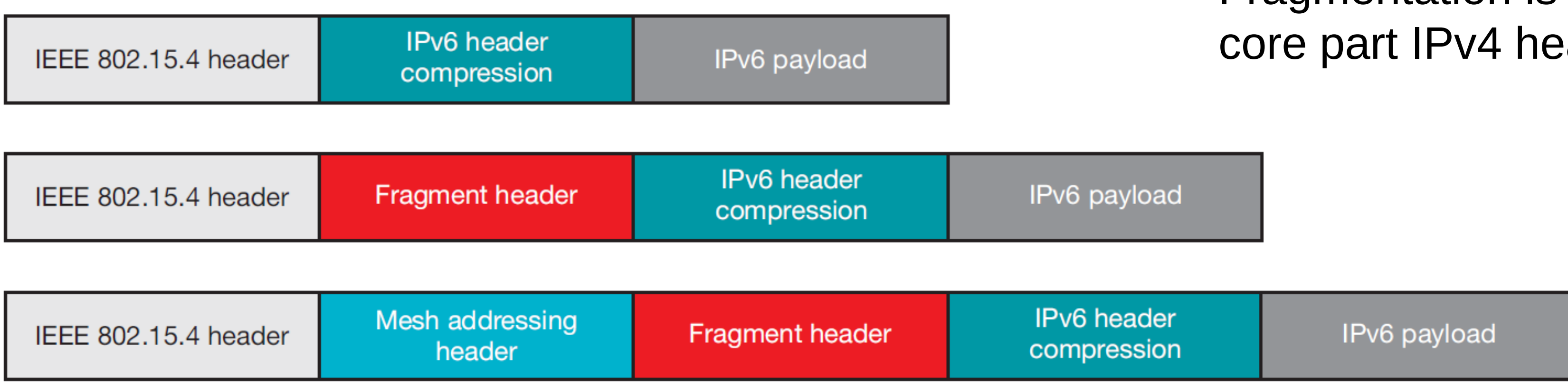
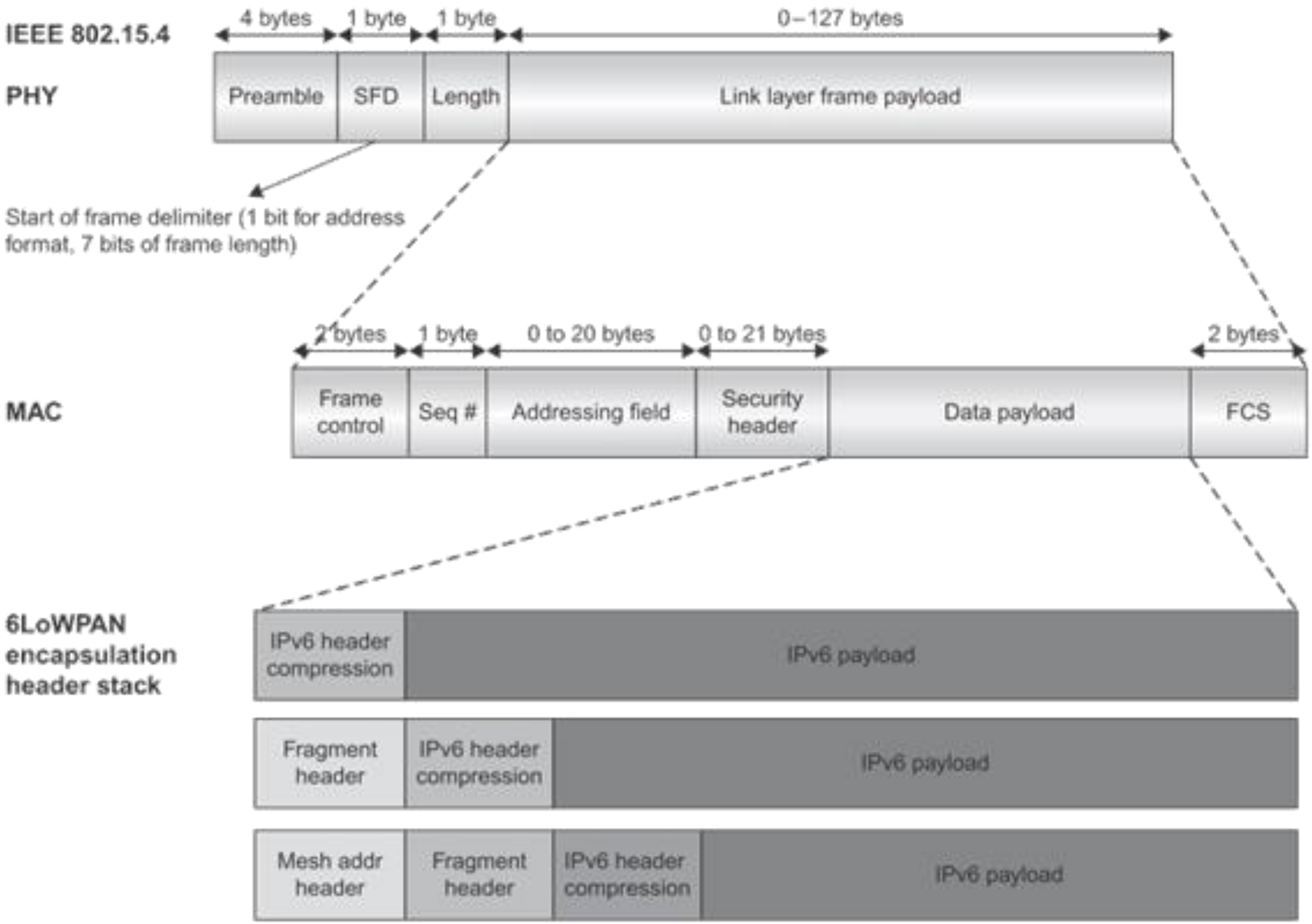


Figure 4. 6LoWPAN stacked headers

LoWPAN Encapsulation



LoWPAN encapsulation are the payload in the IEEE 802.15.4 MAC protocol data unit (PDU).

Cont…

- All LoWPAN encapsulated datagrams are prefixed by encapsulation header stack.
- Each header in the stack starts with a header type field followed by zero or more header fields

```
A LoWPAN encapsulated IPv6 datagram:

    +---------------+-------------+---------+
    | IPv6 Dispatch | IPv6 Header | Payload |
    +---------------+-------------+---------+

A LoWPAN encapsulated LOWPAN_HC1 compressed IPv6 datagram:

    +--------------+------------+---------+
    | HC1 Dispatch | HC1 Header | Payload |
    +--------------+------------+---------+

A LoWPAN encapsulated LOWPAN_HC1 compressed IPv6 datagram that
requires mesh addressing:

    +-----------+-------------+--------------+------------+---------+
    | Mesh Type | Mesh Header | HC1 Dispatch | HC1 Header | Payload |
    +-----------+-------------+--------------+------------+---------+
```

6LoWPAN Headers

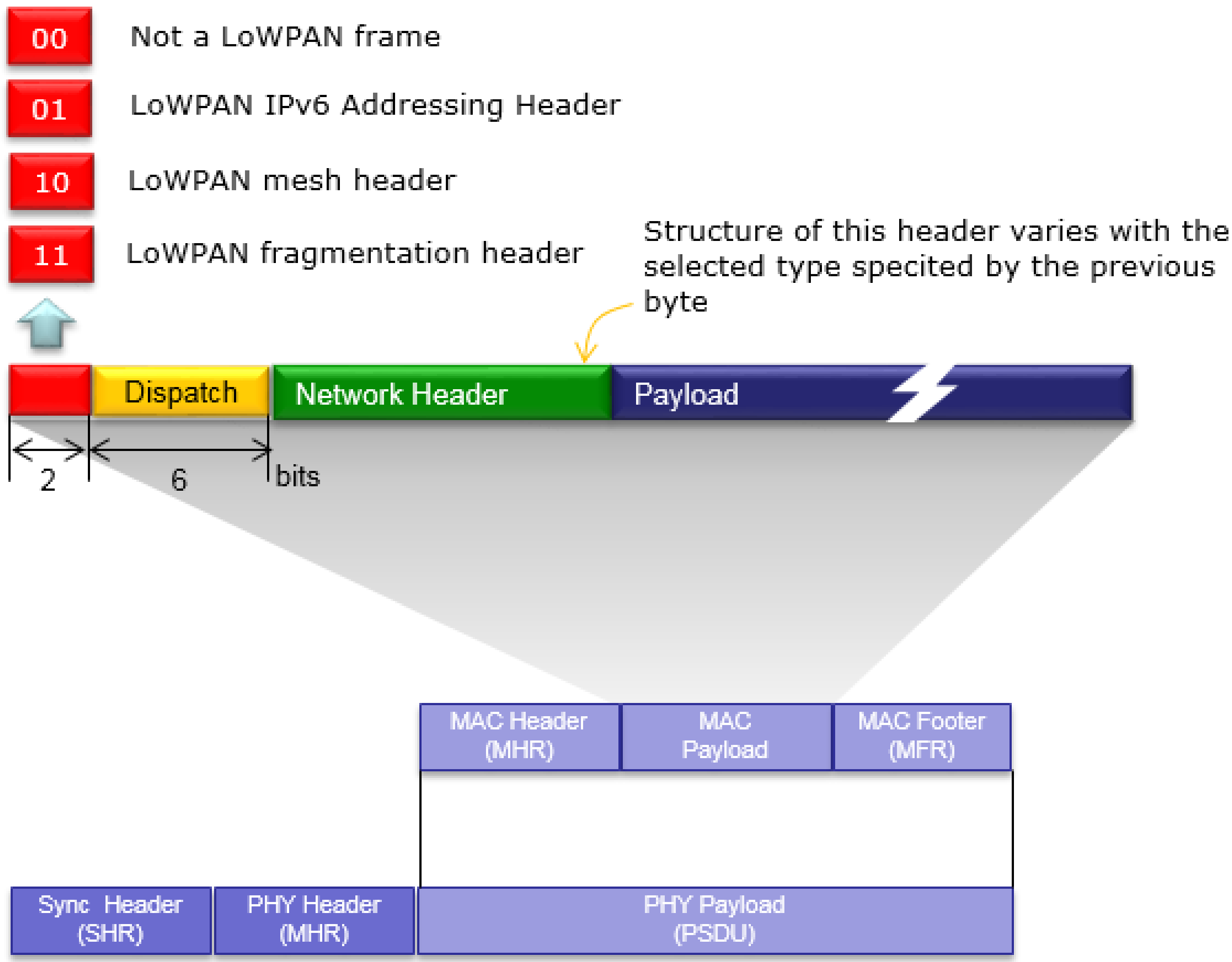
- Each header in the stack starts with
  - a header type field
  - followed by zero or more header fields

- First byte of each header (i.e. dispatch byte) identifies the nature of each header

Bit Pattern	Short Code	Description
00 xxxxxx	NALP	Not A LoWPAN Packet
01 000001	IPv6	uncompressed IPv6 addresses
01 000010	LOWPAN_HC1	HC1 Compressed IPv6 header
01 010000	LOWPAN_BC0	BC0 Broadcast header
01 111111	ESC	Additional Dispatch octet follows
10 xxxxxx	MESH	Mesh routing header
11 000xxx	FRAG1	Fragmentation header (first)
11 100xxx	FRAGN	Fragmentation header (subsequent)

- ✓ NALP -> Not a part of LOWPAN encapsulation, should be discarded.
- ✓ IPv6 -> uncompressed IPv6 header
- ✓ LoWPAN_HC1 -> Compressed IPv6 header.
- ✓ LoWPAN_BC0 -> for mesh broadcast/multicast support header
- ✓ ESC -> supports for dispatch values larger than 127

Cont…



00
Not a LoWPAN frame
01
LoWPAN IPv6 Addressing Header
10
LoWPAN mesh header
11
LoWPAN fragmentation header
Structure of this header varies with the selected type specited by the previous byte
Dispatch
Network Header
Payload
2
6
bits
MAC Header (MHR)
MAC Payload
MAC Footer (MFR)
Sync Header (SHR)
PHY Header (MHR)
PHY Payload (PSDU)

Dispatch Header

Dispatch Type & header

```
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|0 1| Dispatch  |  type-specific header
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

- ✓ Dispatch type defined by 1st & 2nd bits = 01.
- ✓ Dispatch **->** identifies type of header immediately following Dispatch Header
- ✓ Type-specific header which is determined by dispatch header.

Bit Pattern	Short Code	Description
00 xxxxxx	NALP	Not A LoWPAN Packet
01 000001	IPv6	uncompressed IPv6 addresses
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01 111111	ESC	Additional Dispatch octet follows
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11 100xxx	FRAGN	Fragmentation header (subsequent)

Mesh Addressing Header

❑ Mesh Addressing Type & Header

```
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|1 0|V|F|HopsLft| originator address, final address
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

- ✓ 1st & 2nd bits = 10.

- ✓ OA **:** link layer address of originator.
- ✓ FDA **:** link layer address of final-destination

- ✓ V**:** 0 => if OA is 64-bit EUI address
  1 => if OA is16-bit short address

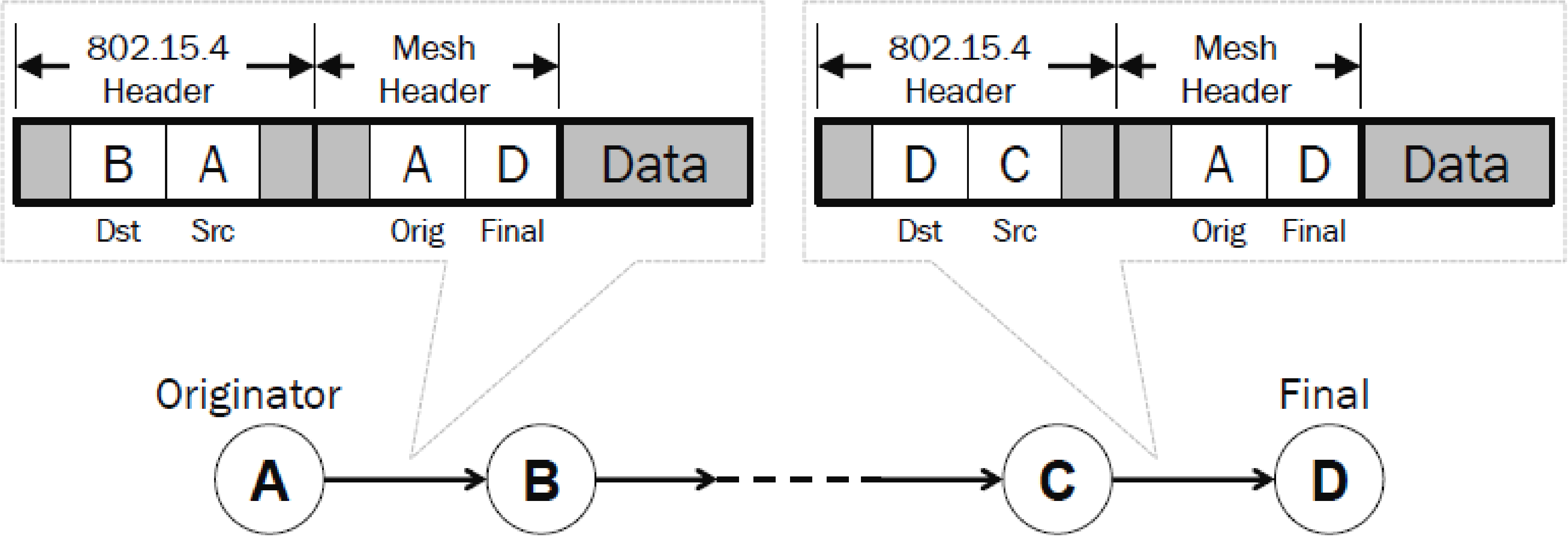
- ✓ F**:** 0 => if FDA is 64-bit EUI address
  1 => if FDA is16-bit short address

- ✓ HopsLft**:** 4 bit, decremented by each forwarding node

Cont…

Example of Mesh Routing header:

- Mesh networking is required to extend the network into multihop scenario



Fragmentation Header

❑ Fragmentation type and Header

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|1 1 0 0 0|    datagram_size    |         datagram_tag          |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
 \_ dispatch _/
```

First Fragment

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|1 1 1 0 0|    datagram_size    |         datagram_tag          |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|datagram_offset|
+-+-+-+-+-+-+-+-+
 \_ dispatch _/
```

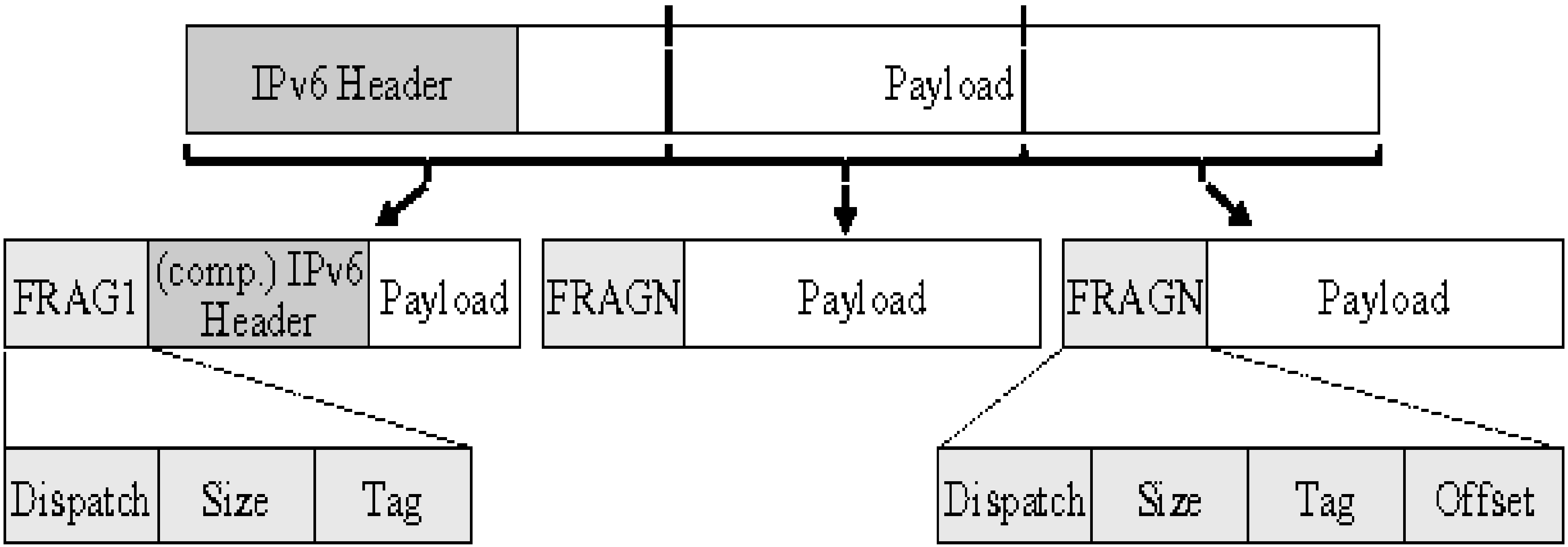
Subsequent Fragments

Datagram size : 11 bit, encode the size of entire packet.
Datagram tag : 16bit, same for all link fragments of a payload.
Datagram offset : present only in subsequent fragments

Fragmentation and Reassembly

Fragmentation Principles (RFC 4944)

- When an IPv6 packet exceeds link-layer payload size then segments the packet into fragments.



6LoWPAN packet structure of FRAG1 and FRAGN

Cont...

- ✓ Only the 1st fragment carries end-to-end routing information.
- ✓ 1st fragment carries a header that includes :
  - datagram size,
  - datagram tag .
- ✓ Subsequent fragments carry
  - the datagram size,
  - the datagram tag,
  - the offset.
- ✓ Time limit for reassembly is 60 seconds.
- ✓ For a lost fragment we need to resend entire set of fragments.

Stateless Address Autoconfiguration

- It defines how to obtain an IPv6 interface identifier (IID)

✓ IPv6 can perform **stateless** auto-configuration of address for an interface.

- Allows a node to connect to the internet without DHCP server
- Uses both local and non-local information to generate its address.

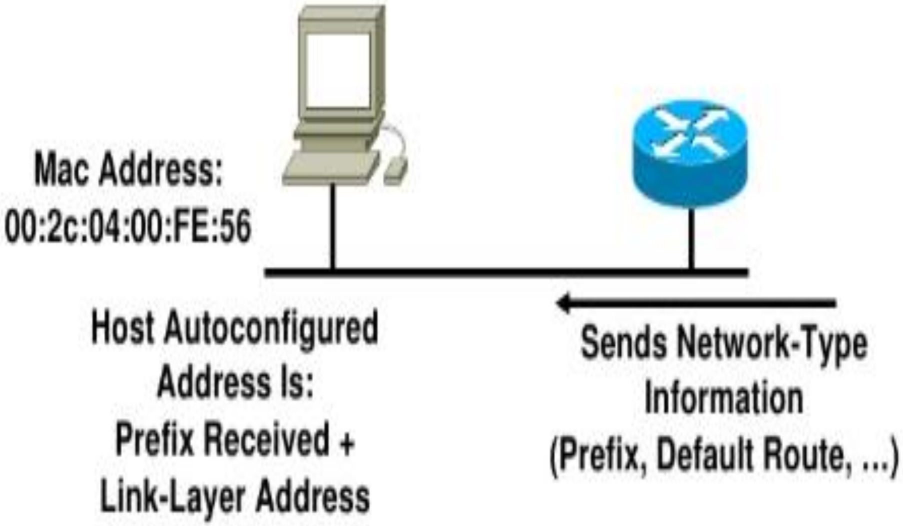
Introduction of IPv6 addressing:

✓ Addresses are 128 bit long

- Divided into 8 hextets, each hextet is 16 bit
- Each character is 4 bit, a nibble
- A common configuration is a
  - 48-bit network prefix,
  - 16-bit subnet mask,
  - 64-bit host address.
    - e.g., 2001:1234:ABCD:0001:1023:FD45:0033:0002

Cont... (for IPv6)

Steps involved in auto-configuration to obtain IPv6 address (128 bits):



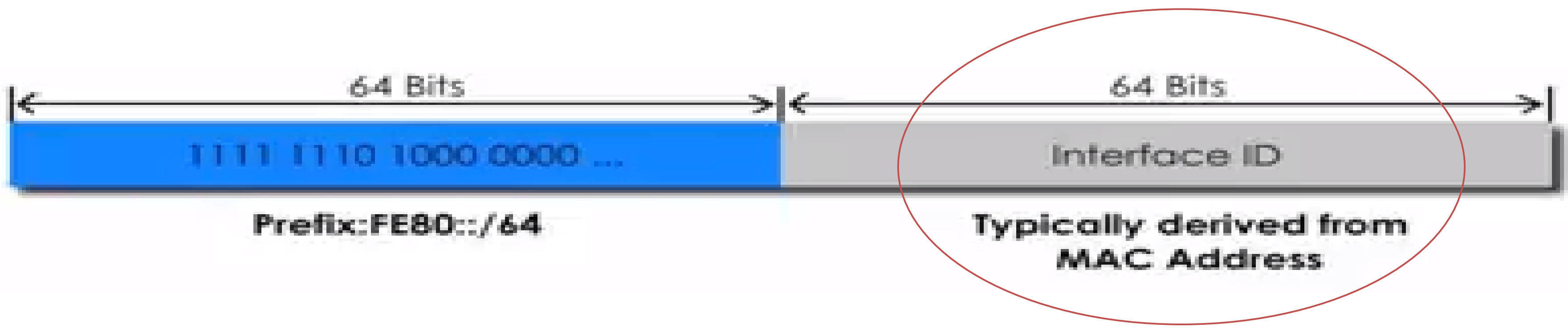
- ✓ **Host** send Router Solicitation (RS) to all routers
- ✓ **Routers** reply with Router Advertisement (RA)
  - ✓ announces prefix used on link.
- ✓ **Host** generates address
  - ✓ by combining the prefix received and host identifier (EUI-64)
- ✓ **Hosts** performs Duplicate Address Detection (DAD)
- ✓ If successful (i.e. not duplicate one), address becomes active
- ✓ Link-local address (128 bit) is generated

Link-local address

A link-local address is a network address that is valid only for communications within the network segment or the broadcast domain that the host is connected to.

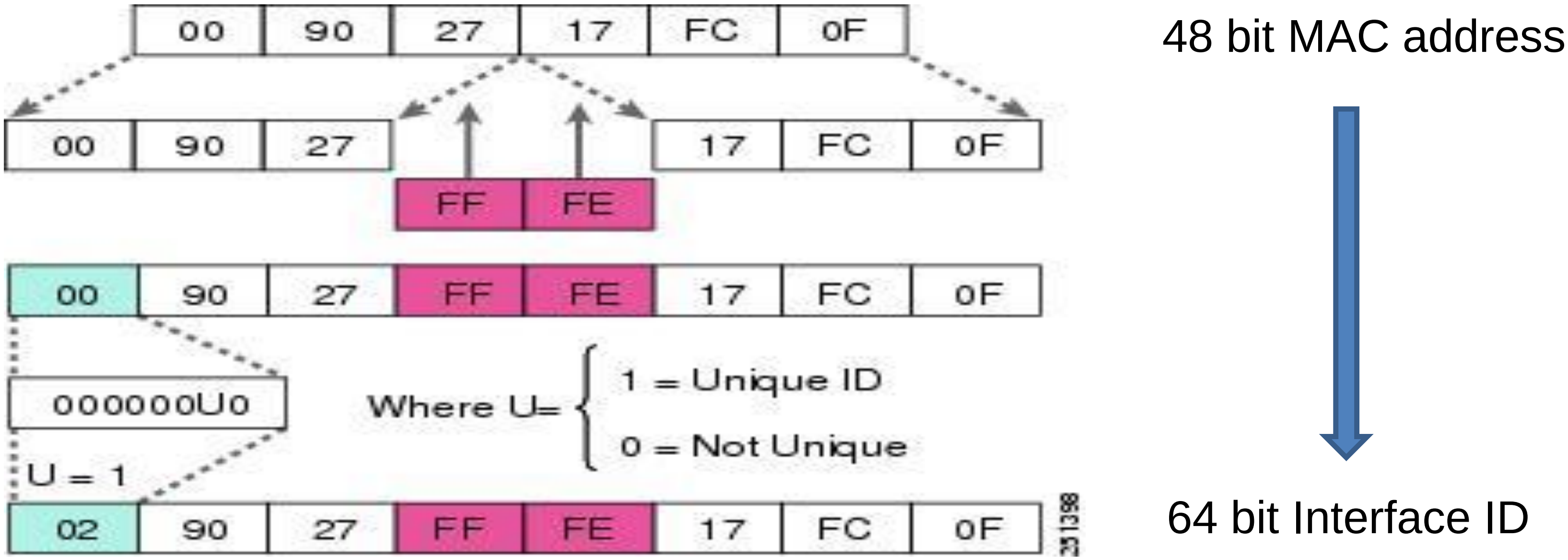
How does a host generate a link-local address in 6LoWPAN?

- ✓ Generated when a computer runs IPv6 boots up
- ✓ Valid only for communication on a local network
- ✓ Always have a prefix FE80::/64



Cont…

Generating IID (EUI-64) from 48-bits MAC address (i.e. device address).



Cont…

All 802.15.4 devices have an IEEE EUI-64 address typically.
But 16-bit short addresses are also possible.

In these cases, a "pseudo 48-bit address" is formed first, and then EUI-64 interface id is formed from the pseudo 48-bit MAC address.

- 1st step: <16 bit PAN ID :16 zero bits>
- 2nd step: <32 bits got in previous step:16 bit short address>
- 3rd step: 48 bits MAC address => 64 bit Interface ID

6LoWPAN Header Compressions

- **Started with HC1 (for IP) and HC2 (for UDP) compressions**
  - Assume common values for header fields and define compact forms.
  - Reduce header size by omission

Omit headers that...

- Can be reconstructed from L2 layer headers (i.e. redundant)
- Contain information not needed or used in the context (i.e. unnecessary)

IPv6 header

Version	Traffic Class	Flow Label (20 bit)	
Payload Length (16 bit)		Next Header	Hop Limit (8 bit)
Source Address			
(128 bit)			
Destination Address			
(128 bit)			

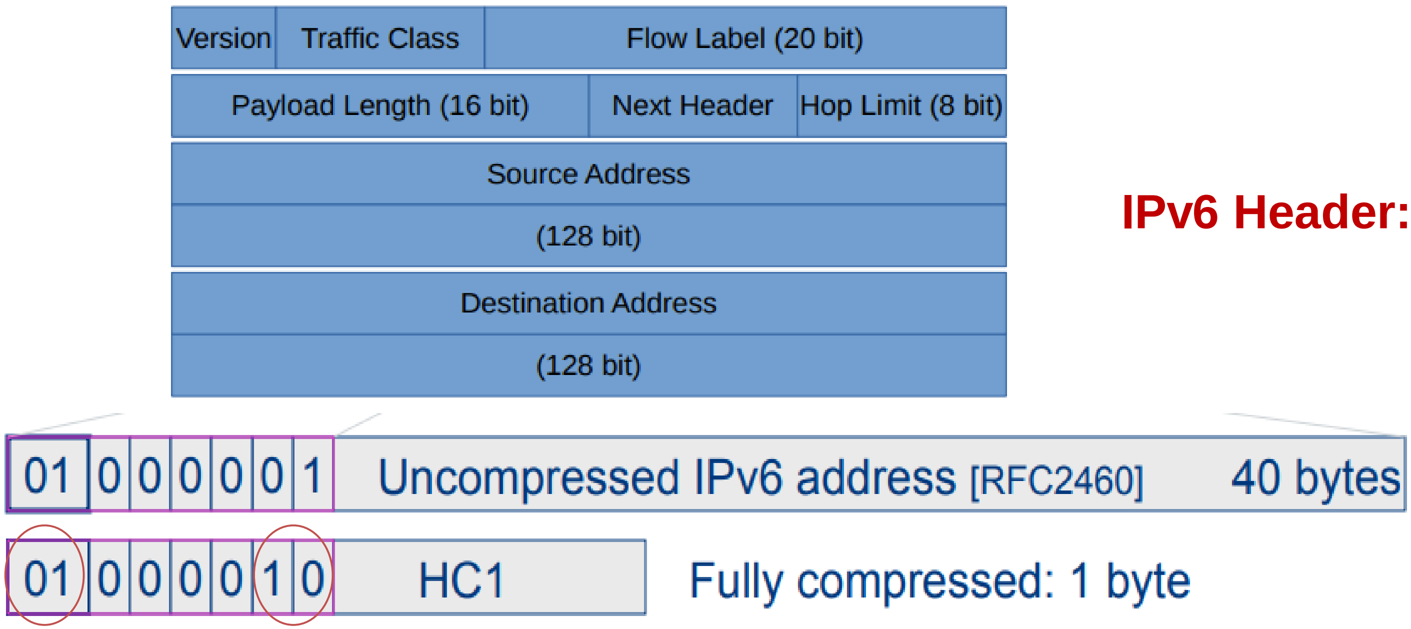
Version: version number of Internet Protocol = 6.
Traffic Class: indicates the class or priority of IPv6 packet

Flow Label: used by source to label the packets belonging to the same flow in order to request special handling by intermediate IPv6 routers
Payload Length: size (in octets) of the rest of the packet that follows the IPv6 header

Next header: type of header that immediately follows the IPv6 header
Hop limit: Decremented by one by each node that forwards the packet.

Compressions: HC1

HC1: Compresses IPv6 headers



IPv6 Header:40-Byte

How to get other fields:

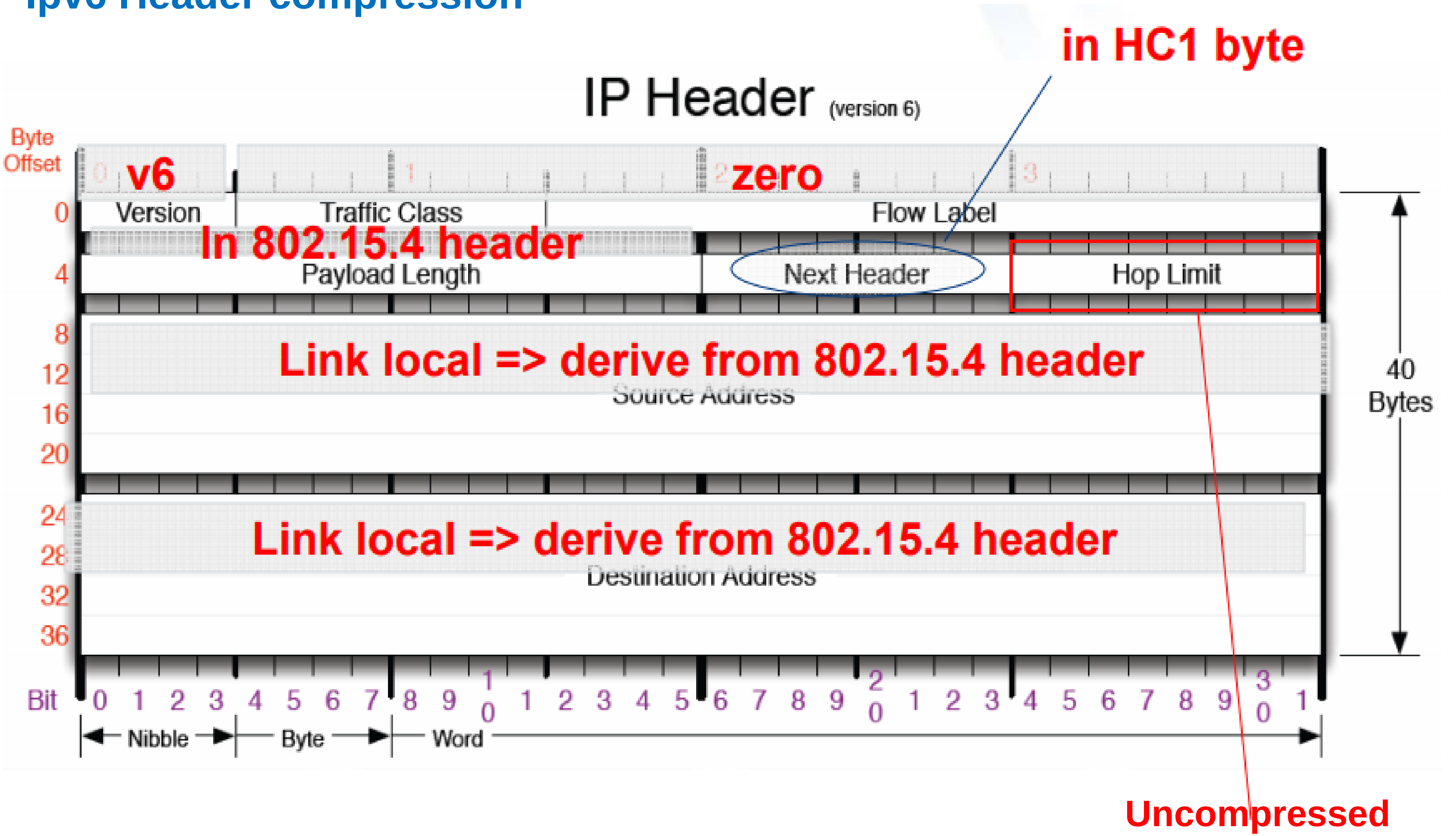
Source address -> **Derived** from link address
Destination address -> **Derived** from link address

Traffic class & Flow level -> Zero (if ECN, DS, Flow level all zero)
Next Header -> Indicated in HC1 (TCP/UDP/ICMPv6)

Reduced to 0 Byte

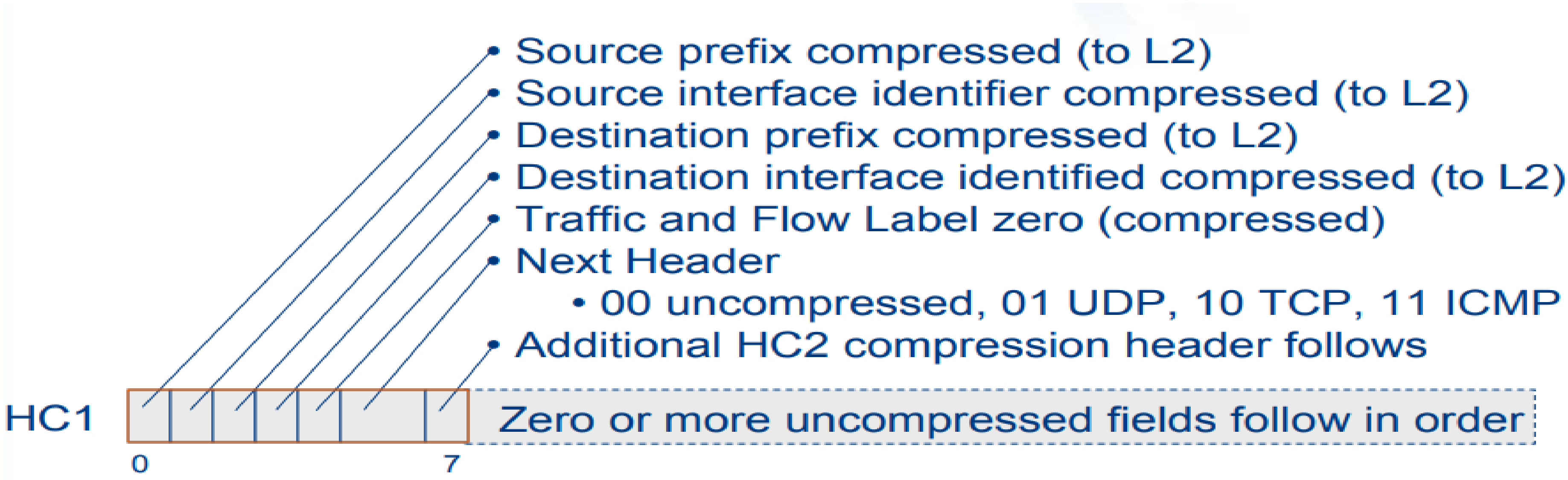
Cont...

Ipv6 Header compression



Cont...

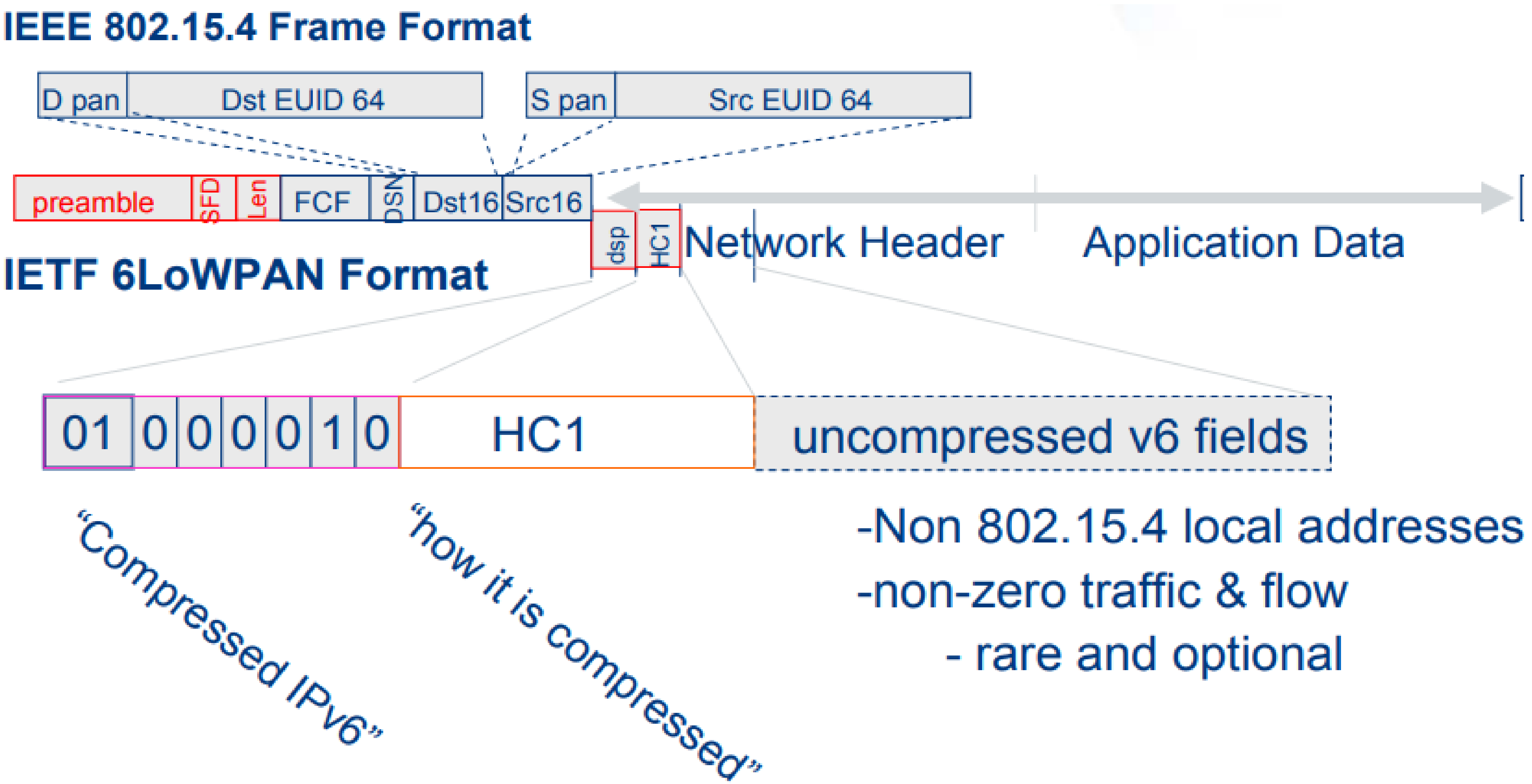
HC1 Compressed IPv6 Header



- ✓ IPv6 address <prefix64 || Interface ID64> for nodes in IEEE 802.15.4 subnet derived from the link address.
  - PAN ID maps to a unique IPv6 prefix
  - Interface ID generated from EUID 64 or PAN ID or short address
- ✓ HopLimit is the only incompressible IPv6 header field.

Cont...

6LoWPAN: Compressed IPv6 Header



-Non 802.15.4 local addresses

-non-zero traffic & flow

- rare and optional

Compressions: HC2

HC2:Compresses UDP Headers

UDP header format
[8 Byte]

Source port	Destination port
UDP length	Checksum

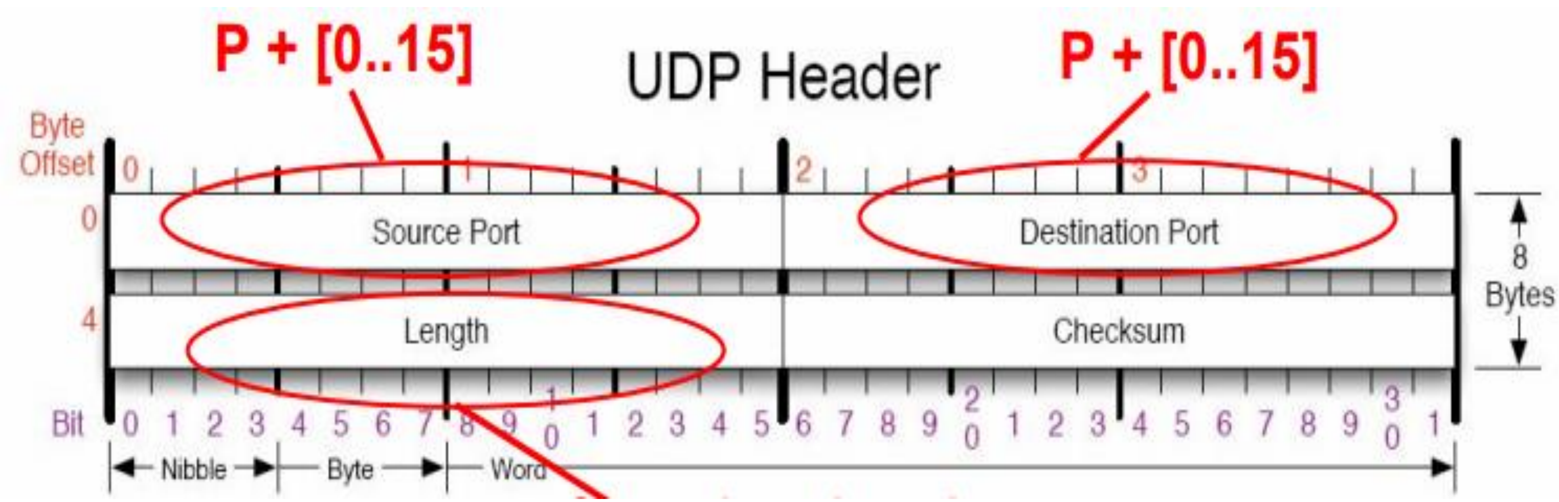
Reduced to

- ✓ Source port = P+4bits, P=61616
- ✓ Destination port= P+4bits, P=61616

1 Byte

- ✓ Length **derived** from IPv6 length
- ✓ Checksum is always carried inline

2 Byte



From 15.4 header

Limitations of HC1&HC2

- ✓ LoWPAN_HC1 & LoWPAN_HC2 are insufficient for most practical uses.
- ✓ Effective only for link-local unicast communication,

- ✓ So, they are usually not used for application layer data traffic in present times

So, RFC 6282 came as an advancement

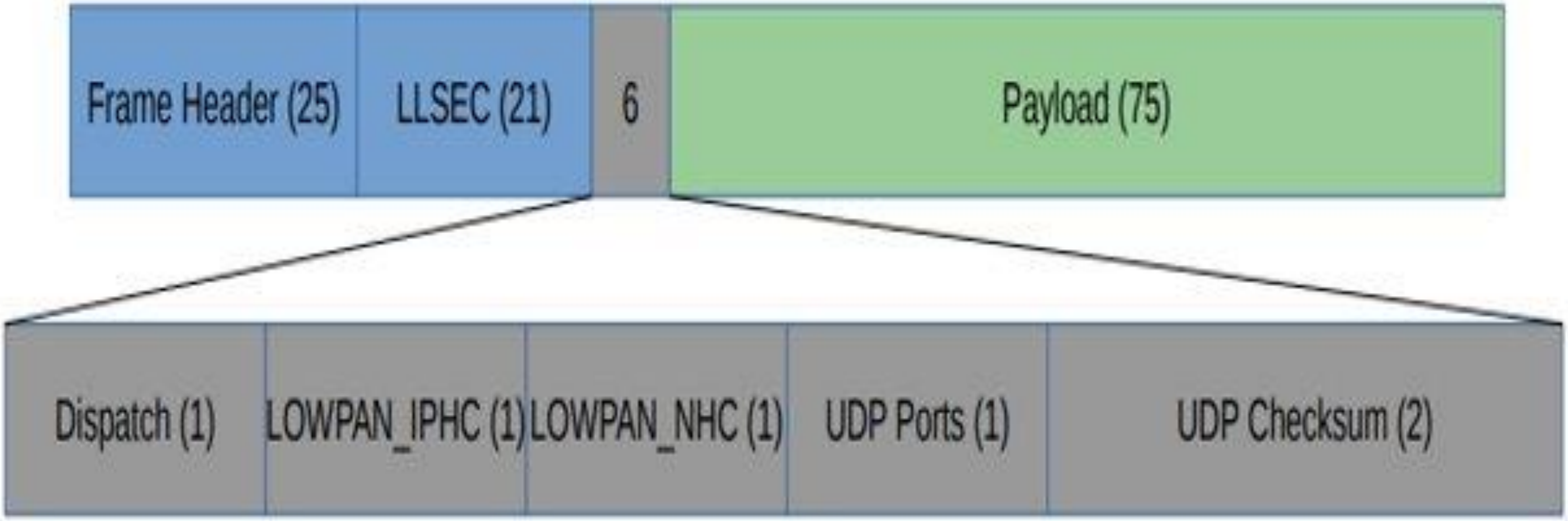
- Defines LoWPAN_IPHC
  - Better compression for global and multicast addresses not only link- local
  - Compress header fields with common values: version, traffic class, flow label, **hop-limit**

- Defines LoWPAN_NHC (for arbitrary next headers)
  - Adds ability to elide UDP checksum

- ✓ Possible to invent your own scheme if you have repeating usage patterns in your use case

The Header Size Solution

The **48-byte (IPv6 + UDP header)** could in the best cases be **reduced to 6 bytes.**



Forwarding Mechanisms

6LoWPAN supports two routing mechanisms:

- **Mesh-under**
- **Route-over**

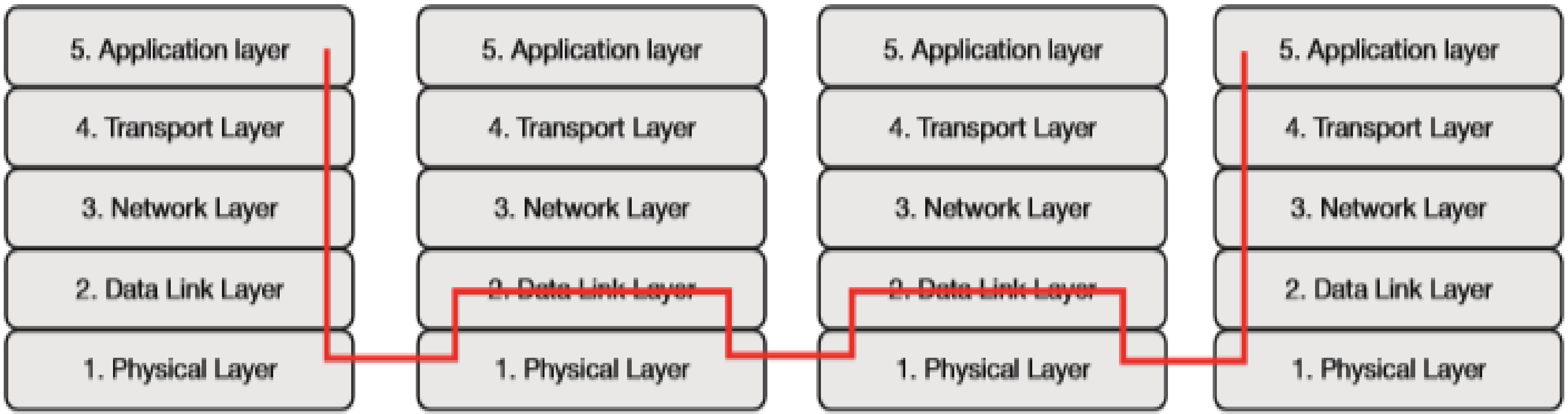
Mesh-under

- ✓ Uses L2 addresses to forward data
- ✓ Only IP router is the edge router
- ✓ Individual fragments may take different paths.
- ✓ Suitable for small and local networks

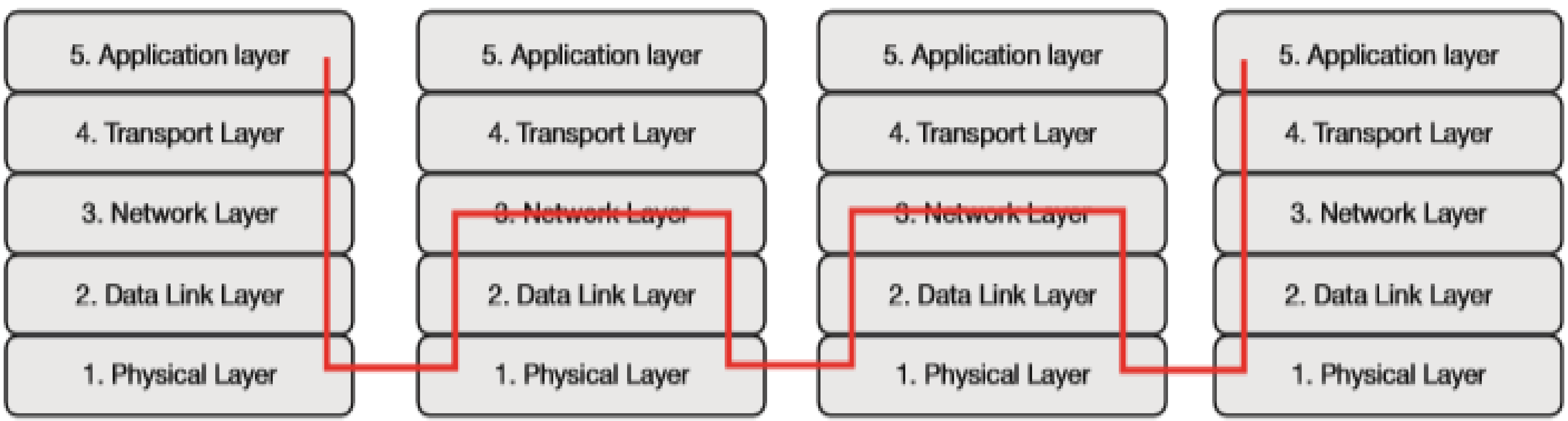
Route-over

- ✓ Uses L3 addresses to forward data.
- ✓ Each hop acts as an IP router.
- ✓ As routing decision taken on a per packet basis, all fragments are sent to same path

Cont…



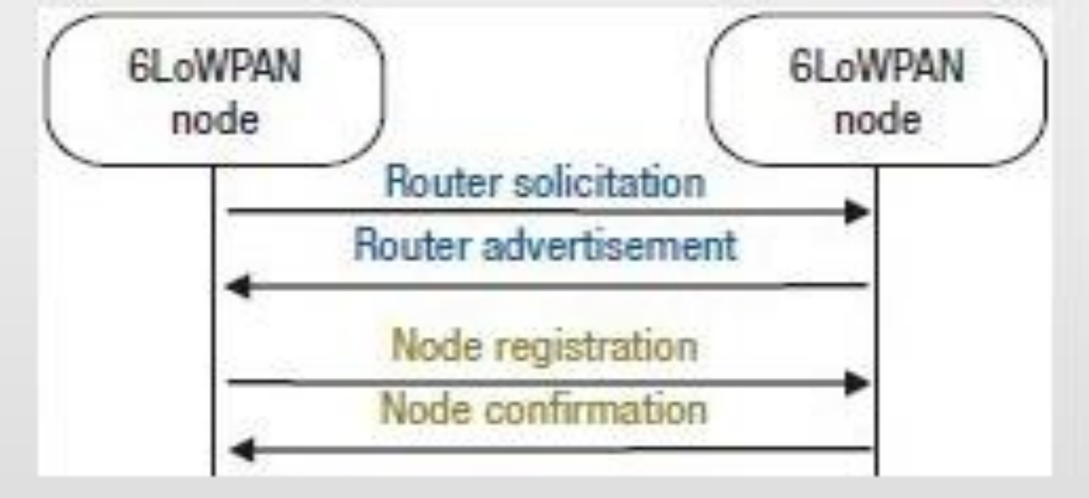
Mesh-under



Route-over

Neighbour Discovery Protocol

- Hosts uses to discover Routers
- Duplicate Address Detection(DAD)
- Uses ICMPv6
  - RS(Router Solicitation) , RA (Router Advertisement)
  - NS(Neighbour Solicitation) , NA (Neighbour Advertisement)



6LoWPAN Neighbour Discovery

Cont...

Router Solicitation and Advertisement

1—ICMP Type = 133 (RS)

Src = link-local address (FE80::1/10)

Dst = all-routers multicast address (FF02::2)

Query = please send RA

2—ICMP Type = 134 (RA)

Src = link-local address (FE80::2/10)

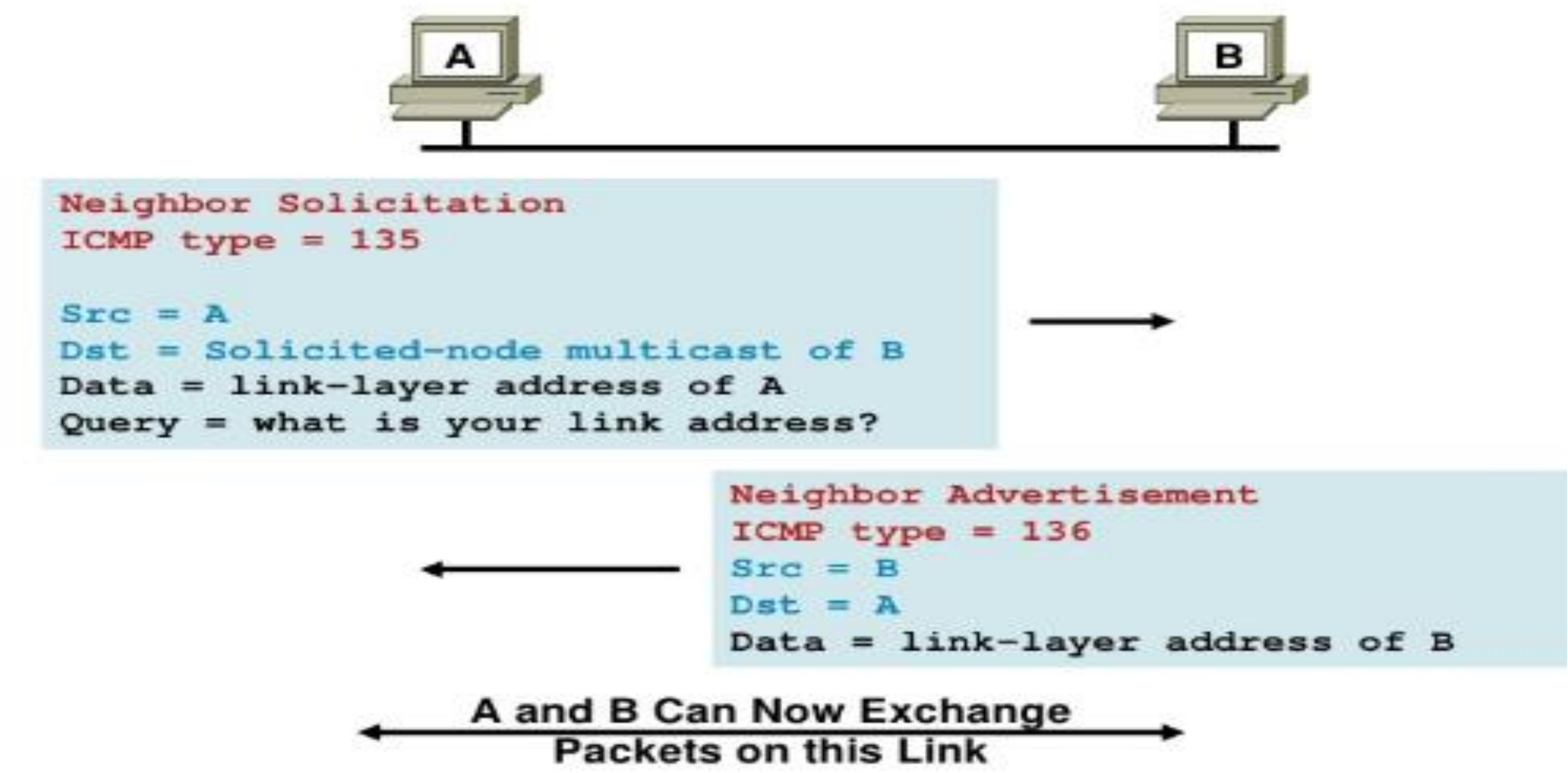
Dst = all-nodes multicast address (FF02::1)

Data = options, subnet prefix, lifetime, autoconfig flag

- Router Solicitations (RS) are sent by booting nodes to request RAs for configuring the interfaces
- Routers send periodic Router Advertisements (RA) to the all-nodes multicast address

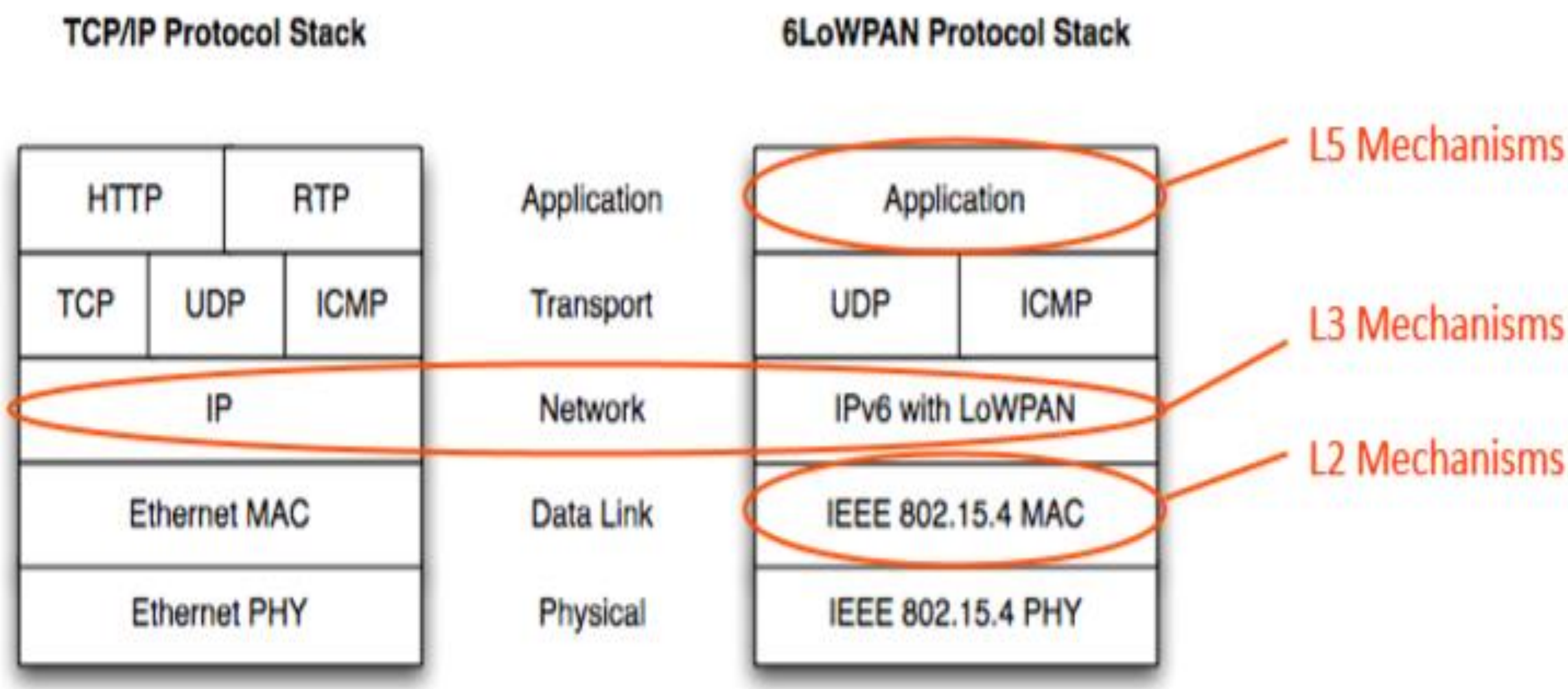
Cont...

Neighbor Solicitation and Advertisement



6LoWPAN Security

- Security is also important for IOT systems
- It takes advantage of 802.15.4 **link layer security**
- Also TLS (**Transport Layer Security**) mechanisms works for 6LoWPAN systems



Lessons Learned

- ✓ What is Open Standard
- ✓ Motivation behind the development of Adaptation Layer
- ✓ 6LoWPAN
  - ✓ Architectures
  - ✓ RFCs
  - ✓ Stacked headers concept
  - ✓ Dispatch Header
  - ✓ Fragmentation Header
  - ✓ Mesh Routing Header
- ✓ Address Autoconfiguration
- ✓ Header Compression
- ✓ Neighbour Discovery
- ✓ Data Forwarding : Mesh under & Route over
- ✓ 6LoWPAN Security

Thanks!